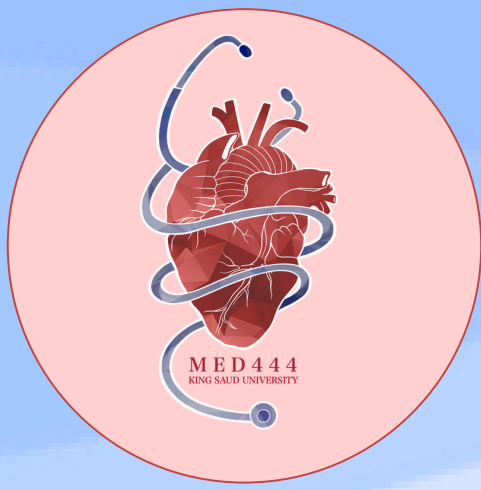




Clinical decision support system



IMPORTANT!

The Stars in this lecture are the important points mentioned by Dr. Samar

Special thanks to Omar Albaqami and Nasser Alshahrani



Editing files

- main text- male slides

- female slides- important

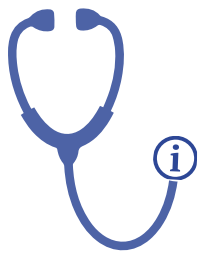
- dr notes - textbook

- extra / old slides

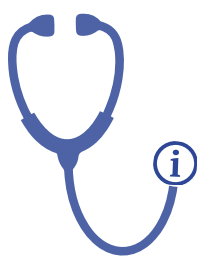
OBJECTIVES



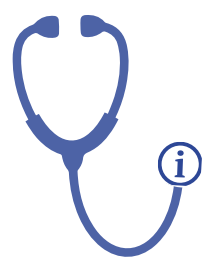
Define electronic clinical decision support (CDS)



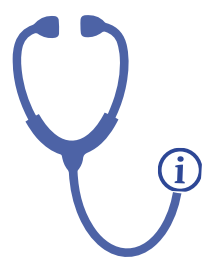
Enumerate the goals and potential benefits of CDS



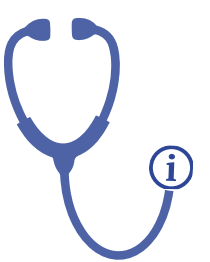
Discuss organizations supporting CDS



Discuss CDS taxonomy, functionality and interoperability



List the challenges associated with CDS

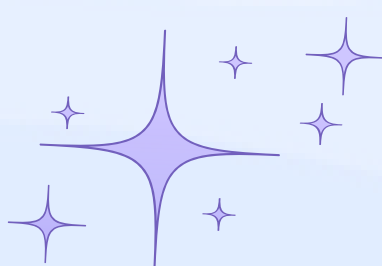


Enumerate CDS implementation steps and lessons learned

The lecture was presented through blackboard by:Dr.samar binkheder

The lecture was presented through blackboard by:Dr.samar binkheder

[Click here to view the resource was attached with slides in blackboard](#)



اللهم انفعني بما علّمتني، وعلمني ما ينفعني، وزدني علماً



Introduction to CDS & CDSS

CDS & CDSS Introduction

- “Clinical decision support (CDS) provides clinicians, staff, patients or other individuals with knowledge and person-specific information, intelligently filtered or presented at appropriate times, to enhance health and health care.”
patient is involved in the decision process. The Office of the National Coordinator for Health IT (ONC).
- Clinical Decision Support System (CDSS)—Information technology systems that support electronic CDS.
- Early on: CDS was thought of **only** in terms of reminders & alerts.
- Now: CDS can include **diagnostic help, cost data** (help the patient make decisions and assess her/his options), **calculators (drug-drug interactions), up-to-date, etc.** (CDSS isn't only about reminders & alerts, they are beyond that)
- Vision: CDS data to be electronic, structured (which helps move the data from system to system easily), and computable (can perform computable tasks and apply technology to it easily).

Though, we can use the Internet's potent search engines **to answer questions**, many organizations promote CDS as a major strategy **to improve patient care and safety**.

Despite extensive online medical resources available to all members of the healthcare team, questions concerning correct diagnoses and optimal treatments still arise frequently. For that reason, many experts have strongly promoted CDS **because it provide you the correct information at the right time.** and it being implemented with EHR is an additional positive.

Five Rights of CDS:



1

Right Information(what) (high quality\evident information from high quality article and clinical guidelines).

2

To the **Right People(who)** (the person **who** is making the clinical decision - physician, patient, or other members of the healthcare team- ex: pharmacist).

3

In **Right Intervention format(how)**(should the information appear as part of an alert, reminder, info button or order set?). That depends on the issue and clinical setting.

4

Through the **Right Channel(where)**(should the information be available as an EHR alert, a text message,email alert, etc.?).

5

At the Right Time(when). (we have to “act” at the right time”)new infor-mation, particularly in the format of an alert should appear early in the order entry process so clinicians are aware of an issue before they complete, e.g. an electronic prescription. This adds to the usabilityof the technology.

INTRODUCTION

Definition

The above definition by Dr. Hayward is widely quoted and general in nature. A more recent definition by the Office of the National Coordinator for HIT states “*Clinical decision support (CDS) provides clinicians, staff, patients or other individuals with knowledge and person-specific information, intelligently filtered or presented at appropriate times, to enhance health and health care.*”¹ To be complete, one could argue that any resource that aides the clinician, other healthcare team members or patients with medical decision making should be considered CDS. For the purposes of this chapter, electronic CDS only will be discussed. CDS, most commonly that is part of electronic health records (EHRs) and computerized physician order entry (CPOE), is discussed in the chapter on EHRs. It should be emphasized that CDS tools potentially assist more than just physicians; nurses, pharmacists, radiologists, patients, etc. have the same need for knowledge management. Also, we define clinical decision support systems (CDSS) as the information technology systems that support electronic CDS.

Early in its evolution CDS was discussed in terms of alerts and reminders to clinicians, but as pointed out in the chapter on EHRs, it should be broadened to include many other tools available to clinicians and others at the point of care. This would include diagnostic help, cost data, up-to-date information about emerging clinical problems such as Zika virus, data from disease registries and so forth. The vision is for all CDS data to be electronic, structured and computable, which is frequently not the case.

Despite extensive online medical resources and robust search engines available to all members of the healthcare team, questions concerning correct diagnoses and optimal treatments still arise frequently.² For that reason, many experts have strongly promoted CDS as a driving force towards improved patient care and safety. Therefore, CDS and quality are closely related.

The Five Rights of CDS

CDS subject matter experts have stressed that for CDS to be effective it must include five rights:

1. The right information (what): should be based on the highest level of evidence possible and adequately

referenced. There should be good internal and external validity; that is, the recommendations are based on high quality studies and they can be applied to a similar population of patients. These issues are discussed in the chapter on evidence-based medicine and clinical practice guidelines.

2. To the right person (who): that is the person who is making the clinical decision, be that the physician, the patient or some other member of the healthcare team.
3. In the right format (how): should the information appear as part of an alert, reminder, info button or order set? That depends on the issue and clinical setting.
4. Through the right channel (where): should the information be available as an EHR alert, a text message, email alert, etc.?
5. At the right time (when) in workflow: new information, particularly in the format of an alert should appear early in the order entry process so clinicians are aware of an issue before they complete, e.g. an electronic prescription. This adds to the usability of the technology.³





Introduction to CDS & CDSS

Historical Prospective

- As early as the 1950s scientists predicted computers would aid medical decision
- CDS programs appeared in the 1970s and were standalone (not integrated into EHR) programs that eventually became inactive. (however, nowadays they tend to make the clinical decision support appears within the EHR)
- HELP (Health Evaluation Through Logical Processing) is a CDS tool that was developed by the University of Utah. In the link you will find full description of this CDS tool and on the side some other CDS tools.
- You can find all of the resources:
http://www.openclinical.org/aisp_help.html

Examples of CDS tools

Examples of CDS tools	Approach (technology) used	Purpose
De Dombal's system Developed by Leeds university in the UK	Bayes theorem (Bayesian probability) (a theoretical or statistical approach)	Make Differential diagnosis for acute abdominal pain
Internist-1	IF-THEN statements (rule-based)	Use observations from patients to Predict diagnoses
MYCIN developed by Dr. Edward (Ted) Shortliffe and others at Stanford University.	Rule-based system	Help in Diagnosis and treatment of infections
SnapDx (Apple iOS) a free mobile app that is a diagnostic CDS for clinicians.	Positive and negative likelihood ratios from medical literature (mathematical)	Diagnosis (App covers about 50 common medical conditions)
Isabel 	Inference engine uses natural language processing and supported by 100,000 documents Uses the free text documents that has signs and symptoms and it will generate a list of integrated coded informations. Signs and symptoms are inputted as free text or imported from the EHR and a diagnostic checklist is generated as a standalone tool or integrated with the EHR (SNOMED-CT coded).	A Diagnosis tool

Isabel Story

Why the name Isabel?

Isabel is not an acronym but the name of the little girl whose illness inspired a medical tool diagnosed to help prevent misdiagnosis.
In 1999, 3 year old Isabel Maude nearly fatally misdiagnosed by her family doctor and hospital when her chickenpox developed, undetected, into Necrotizing fasciitis and Toxic Shock Syndrome. The result of this error was two months in hospital including a month in PICU struggling to survive from multiple organ failure and cardiac arrest. Against all odds, she survived her ordeal and, despite undergoing reconstructive surgery nearly two decades later, she is now an ambitious first class honors graduate pursuing a career in London. Surviving this life-threatening illness has given her a determination to live life to the full - conquering Mount kilimanjaro is just the beginning!

If you are interested you can read the whole story at this link:

<https://www.isabelhealthcare.com/about-isabel-healthcare/isabel-story>





CDS & CDSS



CDS Benefits & Goals



Table 8.1: CDS Benefits and Goals	
Benefits and Goals	Details
Improvement in patient safety	Medication alerts
Improvement in patient care	Improved ordering
Reduction in healthcare costs	Improved patient outcomes
Dissemination of expert knowledge	Better chronic disease management
Management of complex clinical issues	Alerts for critical lab values, drug interactions and allergies
Monitoring clinical details	Improved quality adjusted life years (QALY)
Improvement of population health	Improved diagnostic accuracy
Management of administrative complexity	Fewer duplicate lab tests and images
Support clinical research	Fewer unnecessary tests ordered
	Avoidance of Medicare penalties for readmission for certain conditions
	Fewer medical errors
	Increased use of generic drugs
	Reduced malpractice
	Better utilization of blood products
	Sharing of best evidence
	Education of all staff, students and patients
	Use of clinical practice guidelines, smart forms and order sets
	Interdisciplinary sharing of information
	Case management
	Reminders for preventive services
	Tracking of diseases and referrals
	Identification of high-cost/needs patients
	Mass customized messaging
	Supports coding, authorization, referrals and care management
	Ability to identify prospective research subjects

1

Improvement in patient safety (Medication alerts to avoid medication errors), patient care (Improved patient outcomes), & population health.

2

Reduction in healthcare costs (Fewer duplicate lab tests and images).

3

Management of complex clinical issues (Use of clinical practice guidelines, smart forms and order sets, Interdisciplinary sharing of information).

4

Monitoring clinical details (Reminders for clinical preventive services)

5

Management of administrative complexity (managing referrals and insurance billing).

6

Support clinical research (identify research subjects for clinical trials, cohort and prospective).

7

Dissemination of expert knowledge (evidence based information and education materials on hand).

Supporting Organizations

(Organizations that support CDS (click on the links for more info about the organizations):

Organization	Details
<u>Institute of Medicine (IOM)</u> (now named as National Academy of Medicine (NAM))	They promoted “automated clinical information and CDS” by the use of information technologies to improve access to clinical information.
<u>American Medical Informatics Association (AMIA)</u> Important	Developed 3 pillars of CDS in 2006: 1. Best available evidence. (information is well organized and in correct format when needed during decision making process). 2. High adoption and effective use. 3. Continuous improvement of Knowledge and CDS Methods: both CDS interventions and clinical knowledge undergo continuous improvement based on feedback, experience, and data that are easy to aggregate, assess, and apply.
<u>Office of the National Coordinator (ONC)</u>	Has funded research to promote excellent CDS & sharing possibilities.
<u>Agency for Healthcare Research and Quality (AHRQ)</u>	Also funded multiple CDS research projects and initiatives.
<u>HL7</u> (health level 7). Important	Has a CDS working group and developed FHIR standards, discussed later.
<u>National Quality Forum (NQF)</u>	Developed a CDS taxonomy (help with future quality measurement-CDS is evaluated based on these four components: triggers (how is the performance of the triggers?), input data, intervention, & action steps).
<u>Leapfrog</u>	They have promoted both CPOE and CDS. As part of their approach they developed a CPOE Evaluation Tool that tests a hospital's EHR with multiple mock scenarios such as therapeutic duplications.
<u>Healthcare Information Management System (HIMSS)</u>	Their EMR Adoption Model rates EMRs from 1-7 (Adoption levels). (they provide one of the best books about EMR adoption model) Full use of CDS qualifies as level 6. (in KSA King Faisal Specialist Hospital is the first organization in middle east to achieve level 7)



Types of CDS Methodology



1- Knowledge based CDS, has two phases:

1- Knowledge Use:

Involves the following steps:

1. Trigger (eg: Physician is ordering a medication).
2. Input data (data within EHR like allergies and other medications).
3. Intervention (CDS alert).
4. Action steps (physician takes action either cancel order or continue with the order).

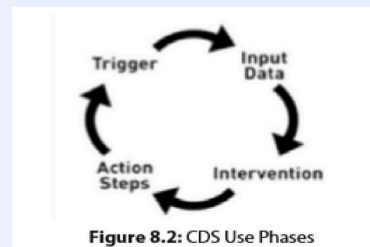


Figure 8.2: CDS Use Phases

2- Knowledge Management: types:

1- Knowledge Acquisition (expert or data):

Expert-based knowledge or data-based knowledge. The former may come from clinical practice guidelines external to the organization and from clinical expertise from within the organization. (Ex. Knowledge that's provided by a physician who's expert in diabetes or knowledge from clinical guidelines because it's written by experts). Data-based knowledge may also come from models built on data from outside the institution (e.g., APACHE scoring) or from data mining from within the institution.

2- Knowledge Representation:

Has many forms. The choice of representation should depend on the problem at hand, the level of expertise of the knowledge engineer, the resources available, and the commitment of the institution to knowledge maintenance. The classical knowledge-based CDS consists of a knowledge base (evidence-based information in right format and time), an inference engine (software to integrate the knowledge with patient-specific data) and a means to communicate the information to the end-user, such as a pop-up alert in the EHR.

3- Knowledge maintenance:

Keep knowledge up to date. This maintenance has proven challenging to most organizations.



Knowledge based CDS = Evidence based, literature based.

Non-knowledge based CDS = AI, machine learning, data mining.

2- Non-Knowledge based CDS

Is Data mining (machine learning) algorithms.

- The previous knowledge representation methods were based on known data so they would be labelled "knowledge based CDS".
- When the knowledge representation is a model derived from data mining, the system is called a non-knowledge based CDS. Data mining methods may involve artificial intelligence (AI), (e.g. neural networks, machine learning) or more traditional statistical methods, like linear or logistic regression. These are data-based systems, that require the models be developed and validated prior to being used in clinical operation. An open-source program for all these methods is the WEKA environment.
- If CDS is based on data mining-related techniques it would be referred to as "non-knowledge based CDS"
- **Advantages of these approaches:**

Analyze large amount of data.

Discovering trends and patterns at the population level. Then, the resulting model can derive recommendations specific to the patient at hand and lie at the heart of predictive analytics.





Types of CDS Methodology



In data mining and machine learning we have two types:	
Supervised learning	Unsupervised learning
- Assumes that the user knows ahead of time what classes or categories exist., eg: this is class o and that class is x. - Types of algorithms used: - Regression model (if the target involves continuous data ((data that can be anything numerical in a range, such as a patient's weight). ex: size of tumors - Classification model (gives us class , dead or alive) (if the target involves discrete ((categorical or nominal)) data ((can only be certain values, such as blood type)). - Types of algorithms used in classification: - Logistic regression binary (two categories-cancer recurrence, no cancer recurrence). - Decision Trees (can be used for the classification and the regression models). There are two types of “decision trees” used in knowledge engineering: - Classification tree, where the attributes of an individual case (patient, population, setting) are input, and the algorithm classifies to one class or the other. - The other type of decision tree is used in decision analysis to derive an optimal action, or event, a flowchart of action. They generally consist of decision nodes (squares), chance nodes (circles) and terminal nodes or outcomes (triangles).Probabilities are assigned to the path branches, while costs or other measures of value or preference are assigned to the outcomes. Mostly used to inform policy, a few decision models have been implemented to provide patients real-time decision support as part of shared decision making. - Neural networks (input data then conclusions of the system then decision based on these conclusions): - Capable of both supervised and unsupervised machine learning. - The results are dependent on a good training set, otherwise the results may not reflect a larger population.	You need to analyse data to get the classes (algorithms try to group similar things together to make classes). a. Clustering (one of the most common ways to analyse large data and group them into distinct groups) (complex method): Most commonly they are grouped as a hierarchy. Applications: - Cluster analysis has been used to group gene sequences, for example identifying clusters of genes associated with breast cancer. - identified discrete groups of chronically ill patients that would benefit from care management (might help in lowering the cost). b. Association rules (IF-THEN) (build relationship between two things, if symptom then diagnosis). have been used for “market basket analysis”, for example, if a man buys diapers, there is a 90% chance he will buy beer.text

سبحانك اللهم إني كنت من الظالمين

لا حول ولا قوة إلا بالله

استغفر الله وأتوب إليه



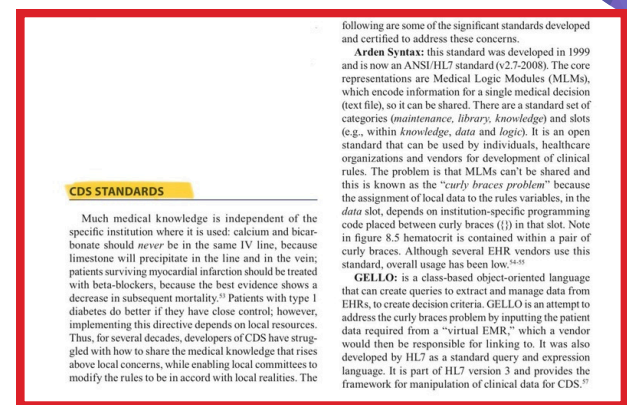


CDS Standards, Sharing & Functionality



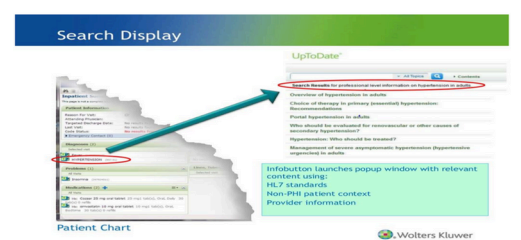
CDS Standards:

- CDS Standards= common language, important for sharing info between institutes.
- CDS developers have struggled for a long time with how to **share knowledge representation** with others or how to modify rules locally.
- Standards were developed to try to overcome **obstacles interoperability** (interoperability means I can move data one from one institute to another and both can understand). (مثلا لو فيه شخصين واحد) (يتكلم عربي والثاني انجليزي ماراح يفهمون بعض عكس لو كانوا كلهم انجليزي)
- Fast Healthcare Interoperability Resources (FHIR):** Developed by HL7 there is great hope that this standard will solve many **Interoperability issues**
 - It is a **RESTful API** (like google uses) that uses either JSON or XML for data representation.
 - It is **data and not document centric**; so a clinician could place a http request on **EHR A** to retrieve just a lab value from **EHR B**. (because it has headings you can locate info and exchange them)
- Infobuttons:** can be placed in workflow where decisions are made with recommendations (links patient's data to general information).



FHIR:

Infobuttons:



CDS Sharing:

There's no single standard that we can guarantee that all hospital use it, each hospital has a different standards



- Currently there is **no single method** for CDS knowledge can be universally shared. The approach has been to either use standards to share the knowledge or use CDS on a shared external server.
- Socratic Grid and Open CDS are open source web services platforms that support CDS.
- The FHIR standard appears to have the greatest chance for success, but it is still early in the CDS game to know

IMPORTANT!



CDS Functionality:

CDSSs can be classified in multiple ways:

Knowledge

- Knowledge based.
- Non-Knowledge based.

Relevance to EHR

- Internal to EHR (functional CDS presents within EHR).
- External to EHR (links take us to another source)

Activation timing

- during patient encounter.
- Before patient encounter.
- After patient encounter.

Alert type

- Interruptive (disruptive): The physician won't be able to do anything until he or she respond to the alert
- Non-interruptive (alerts accumulate)

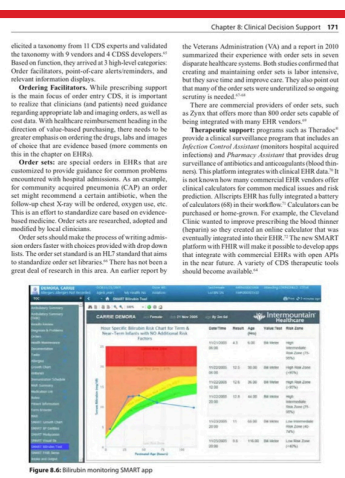
CDS FUNCTIONALITY

There are several other ways to classify CDSS, in addition to knowledge based and non-knowledge-based systems. For example, Osheroff et al. developed a taxonomy of functional intervention types that was related to the type and timing of tasks.³

Importantly, CDS can be located within the EHR or link to a remote CDS program. CDS can activate before, during or after a patient encounter. Activation can be automatic or "on demand." Alerts can be non-interruptive or interruptive (e.g. the process stops until the physician responds to the alert).

Table 8.2 lists a taxonomy of CDS that is based on CDS goals, stated earlier. There is overlap among the categories, as sections such as patient safety and patient care are interrelated.³ 17-18

Wright and colleagues and the Provider Order Entry Team (POET) at Oregon Health & Science University



Smart forms or smart templates are electronic forms that are designed to assist in the collection of data for clinical decision support. They are used to collect data for clinical decision support systems. They are used to collect data for clinical decision support systems. They are used to collect data for clinical decision support systems.

Case Study: Performance of CDS systems in clinical decision support. The study evaluated the performance of CDS systems in clinical decision support. The study evaluated the performance of CDS systems in clinical decision support. The study evaluated the performance of CDS systems in clinical decision support.

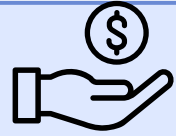
Use of the system related to clinical decision support. The study evaluated the performance of CDS systems in clinical decision support. The study evaluated the performance of CDS systems in clinical decision support. The study evaluated the performance of CDS systems in clinical decision support.



CDS functions (taxonomy)

Taxonomy helps to evaluate CDS system

1. Cost



- Reminders to use generic drugs or formulary recommendations.
- Fewer duplications. (We don't do more tests that we already do, so we reduce the cost).
- Reminders about costs of drugs, lab tests, and imaging studies.
- Reduce Medicare penalties for readmissions
- Reduce medication errors
- Reduce malpractice claims
- Better utilization of blood products

2. Patient safety

- Medication alerts
- Critical lab alerts
- Ventilator support alerts.
- Improved drug ordering for warfarin and glucose
- Infusion pump alerts
- Risk calculation.
- Improved legibility
- Diagnostic aids,

3. Patient care

- Embedded clinical practice guidelines, order sets, and clinical pathways.
- Better chronic disease management.
- Identify gaps in recommended care.
- Immunization aids.
- **Diagnostic aids tools that aid in diagnosis**
- Sepsis alerts. (see case study infobox)
- Antibiotic duration alerts.
- Prognostic aids.
- Patient reminders.
- Pattern recognition for images pulmonary function tests and EKGs, blood gases, pap smear interpretation.

4. Disseminated expert knowledge

- Use of infobuttons for Clinician and patient education (We provide the patient with resources to support him after leaving the clinic)
- Provide evidence based medicine with embedded clinical practice guidelines and order sets.

5. Managing complex clinical issues

- **Reminders for preventative care** for chronic diseases.
- Care management.
- Predictive modeling based on demographics, costs, and clinical parameters.

6. Managing complex administrative issues

- Decision modelling
- Research recruitment

(نظام الفواتير بالمستشفيات)



استغفر الله واتوب اليه



Ordering facilitators

Order sets

Are EHR templated commercial or home grown orders that are modified to follow national practice guidelines .

For example, a patient with a suspected heart attack has orders that automatically include aspirin, oxygen, EKG, etc.

Therapeutic support

include commercial products such as Theradoc and calculators for a variety of medical conditions.

clinical surveillance program that includes an Infection Control Assistant (monitors hospital acquired infections) and Pharmacy Assistant that provides drug surveillance of antibiotics and anticoagulants.

Smart forms

Are templated forms(it has structure question), generally used for specific conditions such as diabetes.

They can include simple check the boxes with evidence based recommendations.

(Advantage: fast filling)

(disadvantage: robot notes- not narrative or complete).

Alerts and reminders

Are the classic CDS output that usually reminds clinicians about drug allergies, drug to drug interactions and preventative medicine and reminders. This is discussed in more detail in the chapter on EHRs and reminders the chapter on patient safety. (Warning for potential problem).



Relevant information delay

لا حول ولا قوة الا بالله

Hyperlinks,
info buttons,
mouse over

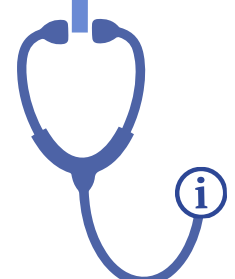
Common methods to connect to evidence based information.
(like mouse over the up to date example)

Diagnostic
support

Most diagnostic support is external and not integrated with the EHR; (e.g. SimuConsult).
Isabel is an example automatically pull coded signs & symptoms.

Dashboards

- Can also be patient, and not population level, so they can summarize a patient's status and thereby summarize and inform the clinician about multiple patient aspects.
 - dashboards can include any type of patient summary, flow chart of lab results or vital signs or a disease registry. (excellent way for data presentation)
- helps the clinician track and trend common medical problems.



CDS Implementation & lessons learned



Implementation is an iterative process

IMPORTANT!

الدكتور نهت انه مهم طريقه الاسئلة ممكن يسأل عن ترتيب
steps او يجيب من التفاصيل اللي من اليمين ويقول في أي
steps تصير

Table 8.3: Logical Steps for CDS Implementation	
Logical Steps	Details
Project initiation	- Ensure clinical and non-clinical leadership are onboard and have a shared vision - Ensure CDS is synched with organizational goals, patient safety/quality measures and meaningful use objectives - Determine the business case/value of CDS for the organization - Determine feasibility from a manpower and financial standpoint and acceptance by clinicians - Ensure objectives are clear and attainable - Identify key stakeholders and assess buy-in - Understand that the CDS needs of specialists are different from primary care - Assess readiness, EHR capability and IT support - Assess the clinical information systems (CIS) involved - Assess knowledge management capabilities - Assemble the CDS team: clinical leaders, CMO, administrative and nursing leaders, managers, EHR vendor and IT experts - Identify clinical champions - Develop CDS charter
Project planning	- Consider a SWOT analysis (strengths, weaknesses, opportunities and threats) - Utilize standard planning tools such as Gantt charts and swim lanes - Develop timeline - Decide whether to build or buy CDS content - CDS committee should select CDS interventions that fit their vision - Be sure to follow the 5 Rights of CDS - Map the different processes involved with CDS and be sure they integrate with the clinician's workflow - Determine whether you will measure structure, processes and/or outcomes - Plan the intervention: triggers, knowledge base, inference engine and communication means - Educate staff and gain their input - Design the CDS program for improvement over baseline performance in an important area for the organization. In other words, be sure you can measure outcomes and compare with baseline data - Investigate the needed CDS standards required - Follow the mandates of change management, e.g. John Kotter's Eight Step Model - Communicate goals of CDS project to all affected
Project execution	- Provide adequate training and make CDS training part of EHR training - Develop use cases - Test and re-test the technology: unit, integration and user acceptance testing - Decide on incremental roll-out or "big bang" - Provide a mechanism for feedback in the CDS process, as well as formal support
Project monitoring and control	- Use data from feedback, override logs, etc. to modify the system as needed - Compare the alert and override rates with national statistics - Measure percent of alerts that accomplished desired goals - Communicate the benefits and challenges to the end-users as they arise - Use tools such as the AHRQ Health IT Evaluation Toolkit - Knowledge management maintenance: are guidelines unambiguous and up to date? Who will maintain the content?

CDS implementation	Details	Lesson learned
Project initiation	- Determine the business) Case/value of CDS for the organization - Asses& institute goals value evision, capabilities and readiness	CDS implementation cannot be a mandate (implementation of CDS is too hard because it needs the Organization to be ready & capable)
Project planning	- Consider SWOT analysis) (strengths weaknesses opportunities and threats - Be sure to follow the 5 Rights of CDS, Educate stafDancKaain their input. تقسيم المتطلبات وكيف راح يكون مره مهم في هذه المرحلة.	One size does not fit all if CDS works in one institution it doesn't mean it will work in another institution
Project execution	-Provide adequate training and make CDS training part of EHR training -Provide a mechanism fo users' feedback) in the CDS process) as well as formal support.	User feedback is critical Very important for improvement of any system implementation
Project monitoring & control	-Use data fromfeedback override logs, etc. to modify th e system as needed.	Knowledge management & up to date content

CDS challenges

Limitations of CDSS are the cost, needs training of health care workers, HCW start relying on machines and forget skills

General

Exploding medical information that is complicated and evolving.
Tough to write rules.

Organisational support

CDS must be supported by leadership, IT and clinical staff
(collaboration the whole team needed) .
Currently, only large healthcare organizations can create robust CDSSs.

Lack of business care

Evidence shows CDS helps improve processes but it is unclear it affects behavior and patient outcomes. Therefore, there may not be a strong business business case to invest in CDSSs
(here we can use CDS taxonomy).

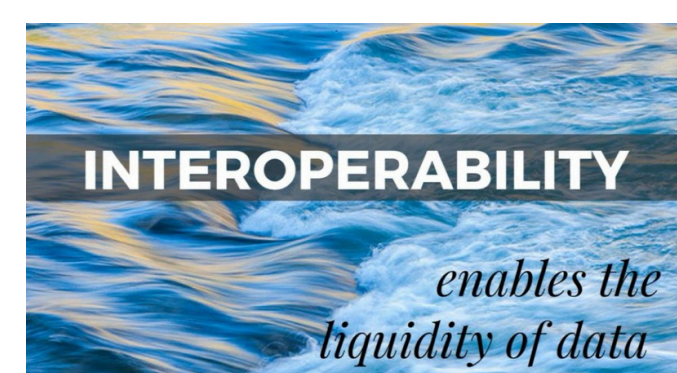


Cont.. CDS challenges



Future trends

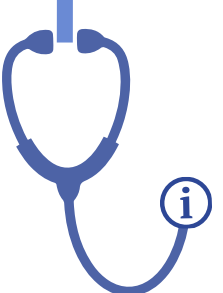
If the FHIR standard makes interoperability easier we may see new CD innovations and improved adoption.

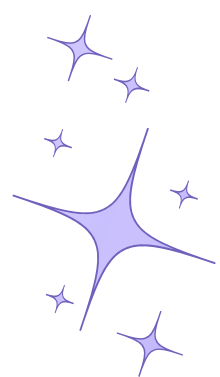


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Conclusion

- CDS could potentially assist with clinical decision making in multiple areas.
- While there is widespread support for CDS, there are a multitude of challenges.
- CDS is primarily achieved by larger healthcare systems.
- The evidence so far suggests that CDS improves patient processes and to a lesser degree clinical outcomes.





SUMMARY



CDS: Clinical decision support system provides clinicians, staff, patients or other individuals with knowledge and person-specific information, intelligently filtered or presented at appropriate times, to enhance health and health care.

Examples

- Internist-1 : IF-THEN statement
- MYCIN: Rule-based system
- Isabel: natural language processing,
- De Dombal's system: Differential diagnosis for acute abdominal pain
- SnapDx (Apple iOS):Diagnosis (App covers about 50 common medical conditions)

Types

Knowledge based CDS:

- knowledge use: trigger > input data > intervention >action steps
- knowledge management

Non-Knowledge based CDS: Data mining (machine learning)algorithm

- supervised learning: regression
- unsupervised learning: clustering

Functions

- **Patient safety:** Medication alerts
- **Cost:** fewer duplication
- **Patient care:** Embedded clinical practice guidelines, Diagnostic aids
- **disseminated expert knowledge:** Use of infobuttons for clinician and patient education
- **Managing complex clinical issues:** Reminders for preventative care for chronic diseases
- **Managing complex administrative issues:** research recruitment

Functionality

ordering facilitators:

- **Order sets:** e.g. Pt with suspected heart attack will automatically include orders of aspirin, oxygen, EKG, etc
- **Therapeutic support:** include calculators
- **Smart forms:** generally for specific conditions
- **Alters and reminders:** remind about drug allergies, drug to drug interactions, and preventatives medicine reminders.

relevant information display:

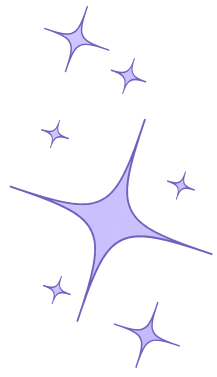
- **Hyperlink, info buttons, mouse over**
- **Diagnostic support:** e.g. ISABEL automatically pull coded signs & symptoms
- **Dashboards:** summarizes patient status and therapy summarize

Implementation

Project initiation > project planning > project execution > project monitoring and control

Challenges

- Organizational support: need all team corporations include IT and clinical staff
- Lack of clear business case: there is still doubt that if CDSS is helping the patients but it shows an effective case regarding processing
- Unintended consequence,
- Medico-legal.
- Must fit Clinical outflow and the 5 rights,
- Technical: require expert IT team,
- Lack of interoperability.
- Require long term commitment



Doctor's questions

Q1: How does a CDSS support evidence-based medicine?

- 1- Patient safety: CDSS strategies are crucial in reducing medication errors, particularly drug-drug interactions (DDIs), which affect up to 65% of inpatients.
- 2-Clinical management: CDSS enhance Adherence to Guideline automated protocols, reminders, and alerts, overcoming traditional Implementation challenges.
- 3-Cost containment: CDSS can reduce costsby decreasing inpatient stays, suggesting alternative medication, and minimizing redundant tests
- 4-Diagnostic support: DDSS systems can assist in diagnosing conditions but mightface challenges like physician suspicion and data gaps

Q2: What are the Clinical Decision Support Systems types, and how do they differ?

- 1- CDSSs have been classified and subdivided into various categories and types, including intervention timing, and whether they have active or passive delivery. CDSS are frequently classified as knowledge- based or non-knowledge based. A) In knowledge-based systems, rules (IF-THEN statements) are created, with the system retrieving data to evaluate the rule, and producing an action or output. Rules can be made using literature-based, practice-based, or patient-directed evidence.
- B) CDSS that are non-knowledge based still require a data source, but the decision leverages artificial intelligence (AI), machine learning (ML), or statistical pattern recognition, rather than being programmed to follow expert medical knowledge
- 2- The Differences are:
Knowledge-based CDSS, and rules are drawn from literature, patient-centered protocols, guidelines, or expert knowledge.
Non-knowledge-based CDSS relies on machine learning or statistical pattern recognition techniques to simulate expert knowledge.

Q3: How can CDSS be used in personalized medicine, and what challenges does it face in this field?

- 1- Using CDSS in personalized medicine:
Clinical Decision Support Systems (CDSS) have significant potential to enable personalized medicine by:
 - Analyzing patient data
 - Improving diagnostic accuracy
 - Optimizing treatment plans
 - Expediting drug trials
- 2- Challenges:
 - Organizational support
 - Lack of a clear business case
 - medico-legal
 - Technical

Q4: What factors should be considered when selecting or developing a CDSS for a healthcare organization?

- 1- Integration of Workflow and User Needs
- 2- Data Interoperability and Quality
- 3- Performance Metrics
- 4- User Training and Support
- 5- Regulatory Compliance

Q5: What are the benefits and drawbacks of using cloud-based CDSS platforms compared to on-premise systems?

On-Premise System: refers to software or hardware that is installed and operated within the physical premises of an organization, as opposed to being hosted remotely or in the cloud.

	cloud-based CDSS platforms	on-premise system
Benefits	1- cost effective 2- scalability 3- accessibility 4- automatic updates 5- collaboration	1- data control 2- customisable 3- faster performance
Drawbacks	1- Data security concerns 2- internet dependency 3- limited customisation	1- higher initial cost 2- limited accessibility 3- maintenance burden

Q6: What are the key considerations for data privacy and security when implementing a CDSS?

- 1-Data privacy:
- 2-Data security:
Educate Staff, Keep Mobile Devices Secure, Encrypt Data, Make the Best Usage of Data Controls, Take Backup



Q7: What are the advantages of non-knowledge-based CDS systems in clinical decision-making?

- 1- simplifies knowledge acquisition & maintenance process
- 2- provide more accuracy in diagnosis disease than average clinician
- 3- can solve complex decision making problems
- 4- useful for research and development in the life sciences space.
- 5- non-knowledge based CDSSs will likely have a more direct role in healthcare and patient engagement.
- 6- Minimizing the instances of prescriptions and medication errors and potential hazardous pairings .
- 7- Analyze large amount of data
- 8- Discovering trends and patterns

Q8: Why is it important to have standardized terminology in CDS systems, and which standards are commonly used?

- 1- The important to have standardized terminology in CDS?
CDS developers have struggled for a long time with how to share knowledge representation with others or how to modify rules locally. Standards were developed to try to overcome obstacles Interoperability Also CDS standards allow CDS tools to be shared, integrated, and implemented in various systems, including electronic health record (EHR) systems.
- 2- Which standards are commonly used ?
FSIR: Fast Healthcare Interoperability Resources is a standard that allows different systems to exchange healthcare data in a standard format. The core of the standard consists of “resources”, which define commonly used healthcare concepts and relationships.
SMART: Substitutable Medical Applications and Reusable Technologies (SMART) is a standard designed to enable the integration of EHR-agnostic medical applications within EHR systems using existing Web standards.





1. Which of the following is NOT one of the “Five Rights” of Clinical Decision Support (CDS)?

A. Right people

B. Right channel

C. Right cost

2. Non-knowledge-based CDS systems primarily rely on which approach?

A. Expert-written IF-THEN rules

B. Data mining and machine learning techniques

C. Manual chart review by clinicians

3. Which standard, developed by HL7, aims to solve many interoperability issues in CDS by using a RESTful API and formats like JSON or XML?

A.leapfrog

B. national quality forum

C. FHIR

4. Which of the following CDS tools does “Isabel” System use?

A. Natural language process

B. Bayes theorem

C. IF-THEN statement

5. What is the correct arrangement of the logical steps?

A. initiation , planning , monitoring and control , execution

B. planning , initiation , execution , monitoring and control

C. initiation , planning , execution , monitoring and control

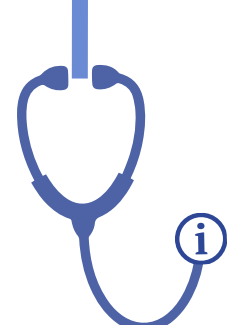
6. (Identify key stakeholders and assess buy-in) Logical Steps of?:

A. Project planning

B. Project initiation

C. Project execution

1-C. 2-B. 3-C. 4-A. 5-C. 6-B





Team Leaders

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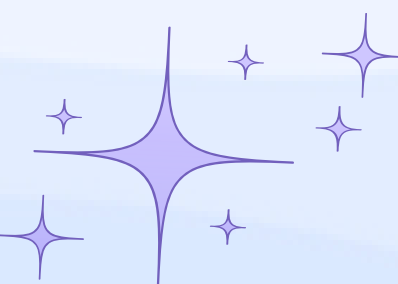


Team Members

Lama Almutair



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سبحان الله، الحمد لله، لا إله إلا الله، الله أكبر،
لا حول ولا قوة إلا بالله

